

Foundation (Jalapeno)

- Q1.** What type of correlation is shown by the data points (2, 4), (4, 6), (6, 8), (8, 10)?
(A) Positive
(B) Negative
(C) None
(D) Inverse
(E) Direct
- Q2.** Which of the following lines of best fit has a positive slope?
(A) $y = -2x + 3$
(B) $y = 2x - 3$
(C) $y = x^2 + 2$
(D) $y = -x^2 - 2$
(E) $y = 0x + 1$
- Q3.** A basketball player scores 10 points in the first game, 12 points in the second game, 14 points in the third game, and so on. If this trend continues, how many points will the player score in the 8th game?
(A) 30
(B) 32
(C) 34
(D) 36
(E) 38
- Q4.** A scatter plot shows the relationship between the times of 10 athletes in a 100-meter dash and their times in a 400-meter dash. If the line of best fit has a slope of 2, what can be concluded about the relationship?
(A) For every 1-second decrease in 100-meter dash time, there is a 2-second decrease in 400-meter dash time.
(B) For every 1-second increase in 100-meter dash time, there is a 2-second increase in 400-meter dash time.
(C) For every 1-second decrease in 100-meter dash time, there is a 1-second decrease in 400-meter dash time.
(D) For every 1-second increase in 100-meter dash time, there is a 1-second decrease in 400-meter dash time.
(E) For every 1-second decrease in 100-meter dash time, there is a 0.5-second decrease in 400-meter dash time.
- Q5.** The table below shows the number of hours spent practicing basketball and the number of three-pointers made in a game. What type of correlation does the data show?
(A) Positive
(B) Negative
(C) None
(D) Inverse
(E) Direct
- Q6.** A runner's times in a 5-kilometer run are 25 minutes, 23 minutes, 21 minutes, 19 minutes, and 17 minutes. If this trend continues, what will be the runner's time in the 8th run?
(A) 5 minutes
(B) 9 minutes
(C) 13 minutes
(D) 15 minutes
(E) 11 minutes
- Q7.** What is the equation of the line of best fit for the data points (1, 2), (2, 3), (3, 4), (4, 5)?
(A) $y = x + 1$
(B) $y = x - 1$
(C) $y = 2x + 1$
(D) $y = 2x - 1$
(E) $y = x^2 + 1$
- Q8.** The scatter plot shows the relationship between the number of hours spent watching sports and the number of sports teams followed. If the line of best fit has a y -intercept of 2, what does this mean in context?
(A) If a person watches 0 hours of sports, they follow 2 sports teams.
(B) If a person watches 2 hours of sports, they follow 0 sports teams.
(C) If a person watches 0 hours of sports, they do not follow any sports teams.
(D) If a person watches 2 hours of sports, they follow 2 sports teams.
(E) If a person watches 0 hours of sports, they follow 0 sports teams.
- Q9.** A swimmer's times in the 100-meter freestyle are 55 seconds, 53 seconds, 51 seconds, 49 seconds, and 47 seconds. If this trend continues, what will be the swimmer's time in the 10th race? Show your work and explain your reasoning.
- Q10.** The table below shows the number of points scored by a basketball team and the number of rebounds in a game. Create a scatter plot and draw a line of best fit. What type of correlation does the data show? Explain your reasoning and provide any necessary calculations.

Open Ended Questions (FRQ)

- Q9.** A swimmer's times in the 100-meter freestyle are 55 seconds, 53 seconds, 51 seconds, 49 seconds, and 47 seconds. If this trend continues, what will be the swimmer's time in the 10th race? Show your work and explain your reasoning.
- Q10.** The table below shows the number of points scored by a basketball team and the number of rebounds in a game. Create a scatter plot and draw a line of best fit. What type of correlation does the data show? Explain your reasoning and provide any necessary calculations.

Chef's Tips

- Tip 1: Remind students to check their units and ensure they are consistent throughout their calculations.
- Tip 2: Encourage students to use graph paper when creating their scatter plots.
- Tip 3: Emphasize the importance of explaining their reasoning and providing clear calculations in the free response questions.